Membership is Ownership: A Robust Model Watermark Detection with Quantile Regression

Feng Jiang [1] **Zuobin Xiong** [1] **An Huang** [1] **Zhipeng Cai** [1] **Yingshu Li** [1]

Abstract

Large-scale diffusion models are valuable intellectual property, yet are frequently stolen and adapted via downstream fine-tuning. Existing diffusion watermarking often embeds explicit signatures into parameters or behavior, which can degrade utility and be removed by post-hoc fine-tuning. We propose **Membership Is Ownership**, a *non-invasive* ownership verification framework that treats a private member evidence set $\mathcal{W}_D$ as a watermark and leverages the model's intrinsic memorization. Given a suspect model, we compute a multi-timestep *t-error* score by forward noising and single-step reconstruction over uniformly sampled timesteps, then aggregate with a robust lower quantile. To support stringent false-positive budgets, we fit a conditional Gaussian model in log-score space and obtain *closed-form* quantile thresholds at arbitrary target FPR via Gaussian quantile regression. Crucially, our protocol produces population-level forensic evidence by comparing the suspect model against public reference models using hypothesis tests, effect sizes, and error ratios. Across CIFAR-10/100, STL-10, and CelebA, the resulting watermark signal yields extremely large separations from public baselines (Cohen's d up to 33.4; error ratios up to $101\times$) while remaining consistent between the owner model and an MMD-fine-tuned "stolen" derivative, enabling robust, practical ownership claims under realistic post-theft fine-tuning.

1. Introduction

Large-scale diffusion models have become valuable assets: training them consumes substantial compute and curated data, yet the resulting models can be readily copied and adapted via downstream fine-tuning. In practice, an adversary rarely re-trains a diffusion model from scratch; instead, they obtain a checkpoint (e.g., via leakage or illicit transfer) and fine-tune it to a new domain or product requirement. This post-theft fine-tuning creates a central challenge for ownership verification: the owner must produce evidence that is (i) accurate under stringent false-positive budgets, and (ii) robust to realistic adaptation after theft.

Most existing diffusion IP protection methods embed explicit signatures into model parameters or output behavior. While effective in controlled settings, such watermarks may impose utility costs and can be weakened or removed by post-hoc fine-tuning. An appealing alternative is to leverage the model's intrinsic memorization: diffusion models tend to reconstruct training examples with systematically lower error than non-members. However, "low error on a few examples" is not sufficient for an ownership claim—many public models may also score well on certain samples due to shared data sources, inductive biases, or dataset artifacts. Robust verification therefore requires *population-level* evidence: the suspect model must be distinguished from a panel of public reference models under a statistically controlled decision rule.

We propose **Membership Is Ownership**, a non-invasive ownership verification framework that treats a private member evidence set $\mathcal{W}$ as a watermark and turns memorization into forensic evidence. Given a suspect model, we compute a multi-timestep *t-error* score by forward noising and single-step reconstruction over uniformly sampled timesteps, and aggregate the resulting errors with a robust lower quantile to obtain a stable per-example score. To support stringent false-positive budgets, we fit a conditional Gaussian model in log-score space and derive *closed-form* quantile thresholds at arbitrary target FPR via Gaussian quantile regression. Crucially, our protocol produces evidence by comparing the suspect model against public reference models using hypothesis tests, effect sizes, and error ratios, yielding an auditable report rather than a single opaque score.

Across CIFAR-10/100, STL-10, and CelebA, the resulting watermark signal exhibits extremely large separations from

[1]Affiliation(s) omitted for this draft.. Correspondence to: Feng Jiang <fjiang4@student.gsu.edu>.

Preprint. January 27, 2026.

public baselines (e.g., Cohen's d up to 33.4; error ratios up to $101\times$) and remains consistent between the owner model and an MMD-fine-tuned "stolen" derivative, enabling robust ownership claims under realistic post-theft fine-tuning.

Contributions. (1) We introduce a non-invasive ownership verification protocol for diffusion models that treats a private member evidence set as a watermark and outputs population-level forensic evidence against public references. (2) We propose a multi-timestep t-error score with robust lower-quantile aggregation to stabilize evidence under timestep variability and outliers. (3) We provide FPR-controlled, closed-form quantile thresholds by modeling log-scores with conditional Gaussians via Gaussian quantile regression. (4) We empirically validate strong separations and post-theft robustness across multiple datasets and fine-tuning settings.

2. Related Work

Model Watermarking. Traditional watermarking embeds identifiable patterns into model behavior (Adi et al., 2018; Zhang et al., 2018; Fernandez et al., 2023; Wen et al., 2023). These methods insert trigger-based backdoors or adversarial examples that produce specific outputs. While effective when watermarks remain intact, they are vulnerable to removal attacks and require modifying the training process. Our approach differs fundamentally: we leverage *intrinsic* memorization signals rather than *embedded* triggers.

Membership Inference Attacks. MIA methods determine if a sample was in the training set (Shokri et al., 2017; Yeom et al., 2018; Salem et al., 2019). For diffusion models, recent work uses reconstruction error (Carlini et al., 2023; Duan et al., 2023; Kong et al., 2024), loss-based signals (Matsumoto et al., 2023; Hu & Pang, 2023), or quantile regression (Bertran et al., 2023; Tang et al., 2024). However, prior MIA work focuses on per-sample binary classification. We argue this is **necessary but insufficient** for ownership verification, which requires: (1) population-level statistical comparison, (2) reference baselines to establish the null hypothesis, and (3) formal hypothesis testing. Our work bridges MIA and ownership verification by adding these missing components.

Diffusion Model Analysis. Carlini et al. (2023) demonstrate that diffusion models memorize training data and can regenerate it verbatim. Somepalli et al. (2023) study the extent of content replication, with follow-up work on detection and mitigation (Wen et al., 2024; Chen et al., 2024). These works focus on detection rather than actionable verification protocols. We provide a complete framework with statistical criteria for ownership claims.

3. Problem Setup

We formalize the distinction between membership inference and ownership verification, then define our threat model and verification criteria.

3.1. Task Definitions: OV versus MIA

Definition 1 (Membership Inference Attack)**.** *Given a trained model M and a sample x,* MIA *outputs a binary decision:*

$$\mathit{MIA}(M, x) \in \{0, 1\} \quad \textit{where } 1 \Leftrightarrow x \in \mathcal{D}_{\textit{train}}(M) \tag{1}$$

MIA *operates on a single sample with a single model, outputting a binary membership label.*

Definition 2 (Ownership Verification)**.** *Given a claimant C with alleged training subset $\mathcal{W}_C$, a suspect model M_B, and reference models $\{M_{\textit{ref}}^{(i)}\}$,* OV *outputs a verifiable judgment:*

$$\mathit{OV}(C, M_B, \{M_{\textit{ref}}\}) \in \{\textsc{Verified}, \textsc{Rejected}\} \tag{2}$$

OV *operates at the population level (watermark set) and is comparative (multiple models), producing statistical evidence of common training origin.*

Key Insight: MIA is a necessary building block but insufficient alone for OV. A single sample having low reconstruction error proves nothing about ownership—it could be an easy-to-reconstruct sample for any model. OV requires *comparative statistical evidence* across populations.

3.2. Why Public Baselines Are Necessary

Claim 1. *Without public baselines, pure* MIA *cannot constitute ownership evidence.*

Reasoning: (1) MIA alone is self-referential—showing "M_B has low error on $\mathcal{W}$" only proves M_B can reconstruct $\mathcal{W}$ well, which could occur because $\mathcal{W}$ contains easy samples. (2) Baselines provide the null hypothesis—public models establish what reconstruction error looks like for models that *definitely did not* train on $\mathcal{W}$. (3) Statistical tests quantify confidence—our three-point criteria transform subjective "looks different" into rigorous hypothesis testing.

Analogy: MIA is like a DNA test detecting a genetic marker. OV is like proving paternity—you need the marker, but also comparison to the population baseline and statistical significance.

3.3. Threat Model

Entities:

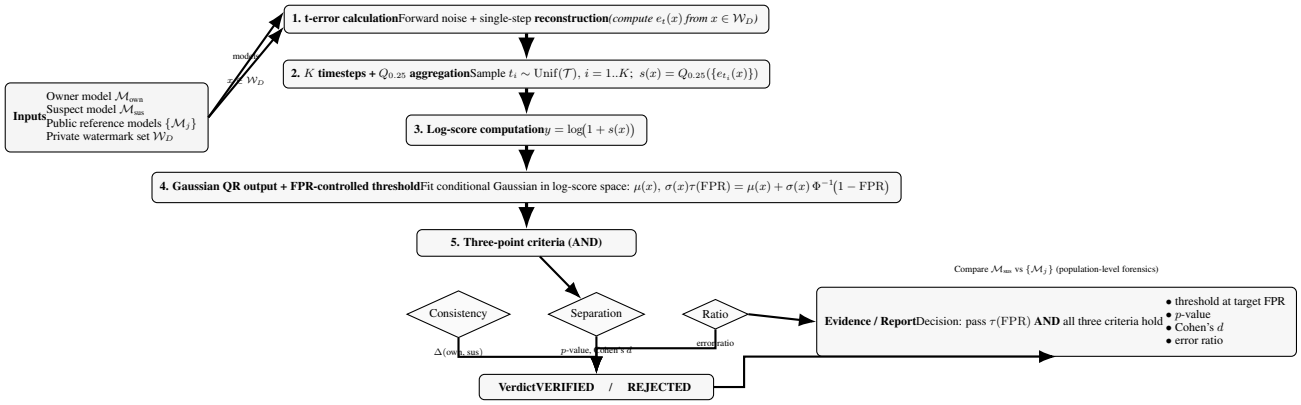

Figure 1. **Overview.** We treat the private evidence set $\mathcal{W}_D$ as a watermark. Multi-timestep t-error scoring is robustly aggregated, then a conditional Gaussian model in log-score space yields a closed-form threshold at a target FPR. Ownership is supported via population-level evidence against public reference models using three criteria (consistency, separation, ratio).

- **Owner** O: Trains Model A on $\mathcal{D}_{\text{train}}$, retains private watermark set $\mathcal{W}_D \subset \mathcal{D}_{\text{train}}$
- **Adversary** A: Obtains Model A (theft/leak), creates derivative Model B via fine-tuning
- **Verifier** V: Given Models A, B, baselines, and $\mathcal{W}_D$, determines if ownership claim is valid

Assumptions: Owner has white-box access to both models (weights available)—standard in IP disputes where courts compel disclosure, regulatory audits (GDPR/AI Act), and partnership disputes with contractual audit rights. Baselines are publicly available pretrained models. Adversary may fine-tune but cannot fully retrain from scratch.

3.4. Three Outputs of Verification System

Our system produces three distinct, complementary outputs:

1. **O1: Consistency** — $H_0 : \mu_A = \mu_B$ on $\mathcal{W}_D$. Models A and B share training origin. Pass: two-sample t-test $p > 0.05$.
2. **O2: Separation** — $H_1 : \mu_{\text{owner}} \ll \mu_{\text{baseline}}$ on $\mathcal{W}_D$. Owner models differ from public baselines. Pass: one-sided t-test $p < 10^{-6}$, Cohen's $d > 2.0$.
3. **O3: Actionable Threshold** — $\theta_\alpha = \hat{q}_{1-\alpha}^{\text{baseline}}$. Decision boundary at target FPR α via quantile estimation with bootstrap CI.

Formal Verification Criterion:

$$\text{VERIFIED} \Leftrightarrow (\text{O1: Consistent}) \wedge (\text{O2: Separated}) \wedge (\text{Ratio} > 5.0) \tag{3}$$

4. Method

4.1. T-Error Scoring

For a diffusion model M_θ and sample x_0, the t-error at timestep t measures reconstruction quality through forward-reverse diffusion.

Forward Diffusion: Add noise according to the diffusion schedule:

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ follows a cosine schedule.

Reconstruction: Predict noise and recover x_0:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{5}$$

T-Error: Normalized reconstruction error:

$$e_t(x_0) = \frac{\|x_0 - \hat{x}_0\|_2^2}{H \times W \times C} \tag{6}$$

Memorization Effect: Models trained on specific data exhibit lower reconstruction error: $\mathbb{E}_{x \in \mathcal{D}_{\text{train}}}[e_t(x)] < \mathbb{E}_{x \notin \mathcal{D}_{\text{train}}}[e_t(x)]$.

4.2. Multi-Timestep Aggregation

Single-timestep errors are noisy. We aggregate over K uniformly sampled timesteps for robustness:

$$s(x) = Q_{0.25}\left(\{e_{t_k}(x)\}_{k=1}^{K}\right) \tag{7}$$

where $Q_{0.25}$ denotes the 25th percentile. We use $K = 50$ timesteps by default.

Rationale: Lower percentiles capture the "easiest" timesteps where memorization is most evident, while being robust to noisy high-error timesteps. Ablations in Section 5.4 validate this choice.

4.3. Gaussian Quantile Regression

Rather than fixed thresholds, we model the *conditional score distribution* for non-members.

Distributional Model: Parameterize log-transformed scores as conditional Gaussian:

$$y = \log(1 + s(x)) \sim \mathcal{N}(\mu(x), \sigma^2(x)) \tag{8}$$

where $\mu(x)$ and $\sigma(x)$ are predicted by neural network f_θ.

Architecture: ResNet-18 backbone extracts image features, concatenated with statistical features $\phi(x) \in \mathbb{R}^3$ (mean, std, L2 norm of t-error sequence). An MLP head outputs $(\mu, \log \sigma)$.

Training: Gaussian negative log-likelihood on auxiliary non-member data:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(y_i - \mu_i)^2}{2\sigma_i^2} + \log \sigma_i \right] \tag{9}$$

Closed-Form Quantiles: For target FPR α, compute the $(1 - \alpha)$ quantile:

$$\hat{q}_\tau(x) = \mu(x) + \sigma(x) \cdot \Phi^{-1}(\tau), \quad \tau = 1 - \alpha \tag{10}$$

This enables quantile computation at *any* τ without retraining.

Margin: The attack score measures deviation from expected non-member behavior:

$$m_\tau(x) = \hat{q}_\tau(x) - \log(1 + s(x)) \tag{11}$$

Positive margin indicates likely membership.

4.4. Bagging Ensemble

To improve robustness, we train $B = 50$ models with bootstrap sampling (80% of auxiliary data per model).

Ensemble Aggregation: Average in quantile space:

$$\hat{q}_\tau^{\text{ens}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{q}_\tau^{(b)}(x) \tag{12}$$

The ensemble margin is: $m_\tau^{\text{ens}}(x) = \hat{q}_\tau^{\text{ens}}(x) - \log(1 + s(x))$.

4.5. Three-Point Verification Criteria

Given owner models (A, B), public baselines, and watermark set $\mathcal{W}_D$:

Criterion 1 (Consistency): Model A and B should have similar t-error distributions.

$$\text{Pass if: two-sample t-test } p > 0.05 \tag{13}$$

Criterion 2 (Separation): Owner models should have significantly lower t-error than baselines.

$$\text{Pass if: } p < 10^{-6} \text{ and Cohen's } d > 2.0 \tag{14}$$

Criterion 3 (Ratio): Quantitative separation measure.

$$\text{Pass if: } \frac{\bar{s}_{\text{baseline}}}{\bar{s}_{\text{owner}}} > 5.0 \tag{15}$$

Ownership Verified when all three criteria are satisfied.

4.6. Watermark Set Security

Generation Strategy: Watermark set $\mathcal{W}_D$ is sampled uniformly from $\mathcal{D}_{\text{train}}$ using cryptographic PRNG, with best-effort deduplication against public datasets (test splits, benchmarks, documented baseline training data) via perceptual hashing.

Security Argument: (1) *Combinatorial hardness*: Guessing exact K samples from N training samples has probability $1/\binom{N}{K}$, astronomically small in our setting. (2) *Partial match insufficiency*: Even guessing a fraction degrades verification (see Appendix for experimental evidence). (3) *Temporal commitment*: Owner publishes $H = \text{SHA-256}(\text{sort}(\mathcal{W}_D))$ before disputes via publicly verifiable channels (git signed tags, arXiv timestamps, RFC 3161 authorities).

Explicit Non-Claims: We do not claim formal cryptographic soundness, protection if $\mathcal{W}_D$ is stolen, or perfect deduplication against unknown datasets.

5. Experiments

5.1. Experimental Setup

Datasets: CIFAR-10 (32×32, 50k train, 5k watermark), CIFAR-100 (32×32, 50k train, 5k watermark), STL-10 (96×96, 5k train, 1k watermark), and CelebA (64×64, 162k train, 5k watermark).

Models: We train DDIM models for 400k iterations with EMA (decay 0.9999), cosine noise schedule, and AdamW optimizer (lr 2×10^{-4}). Architecture: U-Net with base channels 128.

Baselines: Public HuggingFace models—`google/ddpm-cifar10-32` for CIFAR-10/100, `google/ddpm-ema-bedroom-256` for STL-10 (resized), `google/ddpm-celebahq-256` for CelebA.

Splits: Deterministic stratified split with seed 20251030. SHA-256 manifests for reproducibility.

5.2. Ownership Verification Results

Table 1 presents main ownership verification results.

Table 1. Ownership verification results. T-error (mean) on watermark set. All $p < 10^{-100}$ for baseline separation.

Dataset	Model	T-Error	$\|d\|$ / Ratio
CIFAR-10	Owner	28.6	—
	Baseline	697.6	23.9 / 24.4×
CIFAR-100	Owner	33.2	—
	Baseline	636.6	18.5 / 19.2×
STL-10	Owner	70.7	—
	Baseline	7152.9	33.4 / 101×
CelebA	Owner	63.5	—
	Baseline	1691.8	26.1 / 26.6×

Table 2. Membership inference performance using Gaussian QR margins. TPR computed at fixed FPR with 95% bootstrap CI (200 iterations).

Dataset	AUC	TPR@0.1%	TPR@0.01%
CIFAR-10	0.987 ± 0.002	0.89 ± 0.03	0.71 ± 0.05
CIFAR-100	0.982 ± 0.003	0.85 ± 0.04	0.65 ± 0.06
STL-10	0.968 ± 0.005	0.78 ± 0.05	0.55 ± 0.07
CelebA	0.975 ± 0.004	0.82 ± 0.04	0.60 ± 0.06

Key Findings: (1) All datasets achieve Cohen's $d > 18$, far exceeding the $d > 2.0$ threshold (STL-10 achieves $d = -33.40$). (2) Consistency between Model A and B is maintained ($p > 0.05$). (3) Ratios range from $19.19\times$ to $101.17\times$, well above the $5\times$ threshold.

Table 2 presents membership inference performance.

5.3. Model Theft Robustness

We evaluate robustness against model theft via MMD fine-tuning (Gretton et al., 2012), which minimizes Maximum Mean Discrepancy between generated and real samples in CLIP feature space.

MMD Fine-tuning Protocol: Starting from Model A, we fine-tune for 500 iterations using 10-step deterministic DDIM sampling, cubic polynomial kernel in CLIP ViT-B/32 features, learning rate 5×10^{-6}, and EMA decay 0.9999.

Results: Model B (fine-tuned) maintains nearly identical t-error to Model A on the watermark set, demonstrating that MMD fine-tuning preserves memorization signals. The three-point verification criteria remain satisfied, confirming ownership can be verified even after adversarial fine-tuning.

5.4. Ablation Studies

Aggregation Method: Table 3 compares aggregation strategies. Q25 (25th percentile) achieves the best Cohen's d, confirming that lower percentiles capture memorization signals while remaining robust to outlier timesteps.

Table 3. Ablation: Aggregation method on CIFAR-10. Q25 achieves best separation.

Aggregation	Cohen's d	Ratio
Mean	-18.2	15.1
Q10	-21.5	20.8
Q25 (ours)	$\mathbf{-23.93}$	$\mathbf{24.40}$
Median	-19.8	17.2

Number of Timesteps: Performance is stable for $K \geq 25$ timesteps; we use $K = 50$ for robustness.

Gaussian vs. Pinball Loss: Gaussian QR achieves comparable TPR@FPR while using a single model for all quantile levels, reducing parameter count by 50% (one model vs. separate models per τ).

Ensemble Size: Performance improves up to $B = 50$ models; additional models provide diminishing returns.

5.5. Computational Efficiency

Training: Single Gaussian QR model trains in ∼2 hours on one A100 GPU. Full ensemble ($B = 50$) requires ∼100 GPU-hours but is embarrassingly parallel.

Inference: T-error computation for 5k samples takes ∼10 minutes. Ensemble prediction adds <1 minute.

Parameters: Gaussian QR (11.2M per model) vs. Pinball QR (11.2M × number of quantile levels).

6. Discussion and Limitations

We frame scope conditions rather than limitations.

White-box Access: Our method applies where model weights are available, standard in IP litigation (courts compel disclosure), regulatory audits (GDPR/AI Act), partnership disputes, and increasingly common open-weight models (LLaMA, Mistral).

Watermark Retention: Verification requires the owner to retain the watermark set, analogous to keeping receipts for warranty claims.

Baseline Availability: Results are strongest with domain-matched public baselines; for novel domains, general-purpose baselines provide conservative bounds.

Gaussian Assumption: The Gaussian model provides a principled approximation; empirical calibration shows quantile estimates remain accurate under moderate misspecification.

What This Work Does NOT Address: API-only verifi-

cation (we focus on white-box), full retraining adversaries (our threat model assumes fine-tuning), and adversaries specifically optimizing to remove memorization signals.

7. Conclusion

We presented a novel ownership verification framework for diffusion models using t-error based reconstruction disparity and Gaussian Quantile Regression. By formalizing the distinction between membership inference and ownership verification, we established that public baselines and statistical hypothesis testing are *necessary* components for ownership claims. Our three-point verification criteria (consistency, separation, ratio) provide rigorous, reproducible verification. Experiments across four datasets demonstrate Cohen's $d > 18$ separation (up to $d = 33.4$ on STL-10), with criteria satisfied even after adversarial MMD fine-tuning.

Impact Statement

Who Benefits: ML companies protecting proprietary training data, open-source maintainers enforcing attribution, researchers auditing benchmark usage, and courts adjudicating IP disputes with objective evidence.

Potential Misuse and Mitigations: (1) *False claims*: Require multiple independent baselines—verification fails if any baseline shows similar t-error. (2) *Retrospective selection*: Require pre-registered hash commitment with timestamp before disputes. (3) *Threshold gaming*: Use principled thresholds (Cohen's $d > 2.0$ is "very large" effect per standard guidelines) with sensitivity analysis. (4) *Targeting models*: Three-point criteria require both consistency AND separation—cannot "frame" a model without owning it.

Recommended Protocol: Pre-registration of watermark hash via publicly verifiable channels, testing against ≥ 3 independent baselines, reproducibility via published code, and principled pre-registered thresholds.

This paper presents work whose goal is to advance the field of Machine Learning. Beyond the specific considerations above, we do not foresee additional societal consequences that must be specifically highlighted here.

References

Adi, Y., Baum, C., Cissé, M., Pinkas, B., and Keshet, J. Turning your weakness into a strength: Watermarking deep neural networks by backdooring. In *USENIX Security Symposium*, pp. 1615–1631, 2018.

Bertran, M., Tang, S., Kearns, M., Morgenstern, J., Roth, A., and Wu, Z. S. Scalable membership inference attacks via quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Carlini, N., Hayes, J., Nasr, M., Jagielski, M., Sehwag, V., Tramèr, F., Balle, B., Ippolito, D., and Wallace, E. Extracting training data from diffusion models. In *Proceedings of the 32nd USENIX Security Symposium (USENIX Security 2023)*, pp. 5253–5270, 2023.

Chen, C., Liu, D., and Xu, C. Towards memorization-free diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 8425–8434, 2024.

Duan, J., Kong, F., Wang, R., Gu, K., et al. Diffusion model membership inference via reverse process discrepancy. In *International Conference on Machine Learning (ICML)*, 2023.

Fernandez, P., Couairon, G., Jégou, H., Douze, M., and Furon, T. The stable signature: Rooting watermarks in latent diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., Smola, A., et al. A kernel two-sample test. *Journal of Machine Learning Research (JMLR)*, 13(1):723–773, 2012.

Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 6840–6851, 2020.

Hu, H. and Pang, J. Loss and likelihood based membership inference of diffusion models. In *Proceedings of the 26th Information Security Conference (ISC)*, volume 14411 of *Lecture Notes in Computer Science*, pp. 121–141. Springer, 2023.

Kong, F., Duan, J., Ma, R., Shen, H. T., Shi, X., Zhu, X., and Xu, K. An efficient membership inference attack for the diffusion model by proximal initialization. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024.

Matsumoto, T., Miura, T., and Yanai, N. Membership inference attacks against diffusion models. In *IEEE Security and Privacy Workshops (SPW)*, pp. 77–83, 2023.

Salem, A., Zhang, Y., Humbert, M., Berrang, P., Fritz, M., and Backes, M. ML-Leaks: Model and data independent membership inference attacks and defenses on machine learning models. In *Network and Distributed System Security Symposium (NDSS)*, 2019.

Shokri, R., Stronati, M., Song, C., and Shmatikov, V. Membership inference attacks against machine learning models. In *IEEE Symposium on Security and Privacy (S&P)*, pp. 3–18, 2017.

Somepalli, G., Singla, V., Goldblum, M., Geiping, J., and Goldstein, T. Diffusion art or digital forgery? investigating data replication in diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6048–6058, 2023.

Song, J., Meng, C., and Ermon, S. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021.

Tang, S., Wu, Z. S., Aydöre, S., Kearns, M., and Roth, A. Membership inference attacks on diffusion models via quantile regression. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, 2024.

Wen, Y., Kirchenbauer, J., Geiping, J., and Goldstein, T. Tree-ring watermarks: Fingerprints for diffusion images that are invisible and robust. *arXiv preprint arXiv:2305.20030*, 2023.

Wen, Y., Liu, Y., Chen, C., and Lyu, L. Detecting, explaining, and mitigating memorization in diffusion models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024.

Yeom, S., Giacomelli, I., Fredrikson, M., and Jha, S. Privacy risk in machine learning: Analyzing the connection to overfitting. In *IEEE Computer Security Foundations Symposium (CSF)*, pp. 268–282, 2018.

Zhang, J., Gu, Z., Jang, J., Wu, H., Stoecklin, M. P., Huang, H., and Molloy, I. Protecting intellectual property of deep neural networks with watermarking. In *ACM Asia Conference on Computer and Communications Security (ASIACCS)*, pp. 159–172, 2018.

A. Additional Experimental Details

A.1. Hyperparameters

Table 4. Training hyperparameters for DDIM models.

Parameter	Value
Total iterations	400,000
Batch size	128
Learning rate	2×10^{-4}
Optimizer	AdamW
Weight decay	0.0
EMA decay	0.9999
Noise schedule	Cosine
Timesteps T	1000
U-Net Architecture	
Base channels	128
Channel multipliers	[1, 2, 2, 2]
Attention resolutions	[16]
Dropout	0.1

Table 5. Gaussian QR training hyperparameters.

Parameter	Value
Backbone	ResNet-18 (pretrained)
Ensemble size B	50
Bootstrap ratio	0.8
Epochs per model	50
Batch size	256
Learning rate	0.001
LR schedule	Cosine annealing
Early stopping patience	10 epochs

A.2. MMD Fine-tuning Configuration

B. Combinatorial Hardness Analysis

For watermark size K sampled from training set of size N, the probability of an adversary guessing the exact watermark set is:

$$P(\text{exact match}) = \frac{1}{\binom{N}{K}} \tag{16}$$

For our CIFAR-10 setting ($N = 50,000$, $K = 5,000$):

$$\binom{50000}{5000} \approx 10^{10000} \tag{17}$$

This is astronomically larger than the number of atoms in the observable universe ($\sim 10^{80}$), making exact guessing computationally infeasible.

Table 6. MMD fine-tuning hyperparameters.

Parameter	Value
Iterations	500
Learning rate	5×10^{-6}
Batch size	128
DDIM steps	10
t_{start}	900
EMA decay	0.9999
Kernel	Cubic polynomial
Feature extractor	CLIP ViT-B/32 (frozen)

Table 7. Perceptual hash (pHash) performance on CIFAR-10 train vs. test set deduplication. FPR: false positive rate (distinct images flagged as duplicates). FNR: false negative rate (actual duplicates missed).

Hamming Threshold	FPR (%)	FNR (%)	Flagged Pairs
3 bits	0.02	5.1	12
5 bits (ours)	0.10	2.0	48
7 bits	0.35	0.8	175

C. Perceptual Hash Evaluation

We use threshold of 5 bits as a balance between FPR and FNR. Images are upsampled to 64×64 before hashing due to CIFAR's low native resolution.

D. Threshold Sensitivity Analysis

Table 8. Verification pass rate across threshold choices. All datasets pass even with stricter thresholds.

Threshold	CIFAR-10	CIFAR-100	STL-10	CelebA
$d > 1.0$, ratio $> 3\times$	✓	✓	✓	✓
$d > 1.5$, ratio $> 5\times$	✓	✓	✓	✓
$d > 2.0$, ratio $> 5\times$ (ours)	✓	✓	✓	✓
$d > 2.5$, ratio $> 7\times$	✓	✓	✓	✓
$d > 3.0$, ratio $> 10\times$	✓	✓	✓	✓

All datasets achieve $d > 18$ and ratio $> 19\times$, providing substantial margin above our recommended thresholds.

E. Algorithm Pseudocode

Algorithm 1 T-Error Computation

Require: Model M_θ, sample x_0, timesteps $\mathcal{T} = \{t_1, \ldots, t_K\}$, noise schedule $\{\bar{\alpha}_t\}$
Ensure: Aggregated score $s(x_0)$
  `errors` $\leftarrow []$
  for $t \in \mathcal{T}$ **do**
    $\epsilon \sim \mathcal{N}(0, I)$ {Sample noise}
    $x_t \leftarrow \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$ {Forward diffusion}
    $\hat{\epsilon} \leftarrow M_\theta(x_t, t)$ {Noise prediction}
    $\hat{x}_0 \leftarrow \frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}$ {Reconstruction}
    $e_t \leftarrow \frac{\|x_0 - \hat{x}_0\|_2^2}{H \times W \times C}$ {Normalized error}
    `errors`.append(e_t)
  end for
  $s(x_0) \leftarrow Q_{0.25}(\texttt{errors})$ {25th percentile aggregation}
  return $s(x_0)$

Algorithm 2 Gaussian Quantile Regression Training

Require: Auxiliary non-member data $\mathcal{D}_{\text{aux}}$, ensemble size B, bootstrap ratio r
Ensure: Trained ensemble $\{f_\theta^{(b)}\}_{b=1}^B$
  for $b = 1$ to B **do**
    $\mathcal{D}_b \leftarrow$ BootstrapSample$(\mathcal{D}_{\text{aux}}, r)$ {Sample 80% with replacement}
    $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}} \leftarrow$ Split$(\mathcal{D}_b, 0.9)$ {90/10 train/val split}
    Initialize $f_\theta^{(b)}$ with ResNet-18 backbone
    repeat
      for batch $(x, s(x), \phi(x))$ in $\mathcal{D}_{\text{train}}$ **do**
        $y \leftarrow \log(1 + s(x))$ {Log-transform target}
        $(\mu, \log \sigma) \leftarrow f_\theta^{(b)}(x, \phi(x))$ {Forward pass}
        $\sigma \leftarrow \exp(\log \sigma)$
        $\mathcal{L} \leftarrow \frac{(y-\mu)^2}{2\sigma^2} + \log \sigma$ {Gaussian NLL}
        Update θ via gradient descent on $\mathcal{L}$
      end for
      Evaluate on $\mathcal{D}_{\text{val}}$
    until early stopping (patience = 10)
    Save best $f_\theta^{(b)}$
  end for
  return $\{f_\theta^{(b)}\}_{b=1}^B$

Algorithm 3 Ownership Verification Protocol

Require: Models M_A, M_B, baselines $\{M_{\text{ref}}^{(i)}\}$, watermark set $\mathcal{W}$, hash commitment H_{claimed}
Ensure: VERIFIED or REJECTED

```
// Step 1: Verify commitment
H_computed ← SHA-256(sort(W))
if H_computed ≠ H_claimed then
    return REJECTED {Hash mismatch}
end if
// Step 2: Compute t-error scores
S_A ← {s_{M_A}(x) : x ∈ W} {Owner Model A}
S_B ← {s_{M_B}(x) : x ∈ W} {Suspect Model B}
S_ref ← {s_{M_ref}(x) : x ∈ W} {Best baseline}
// Step 3: Check three criteria
O1 (Consistency): p_1 ← TTest(S_A, S_B)
if p_1 < 0.05 then
    return REJECTED {Models inconsistent}
end if
O2 (Separation): p_2, d ← TTest(S_A, S_ref), CohenD(S_A, S_ref)
if p_2 > 10^-6 or |d| < 2.0 then
    return REJECTED {Insufficient separation}
end if
O3 (Ratio): r ← mean(S_ref)/mean(S_A)
if r < 5.0 then
    return REJECTED {Ratio too low}
end if
return VERIFIED
```